

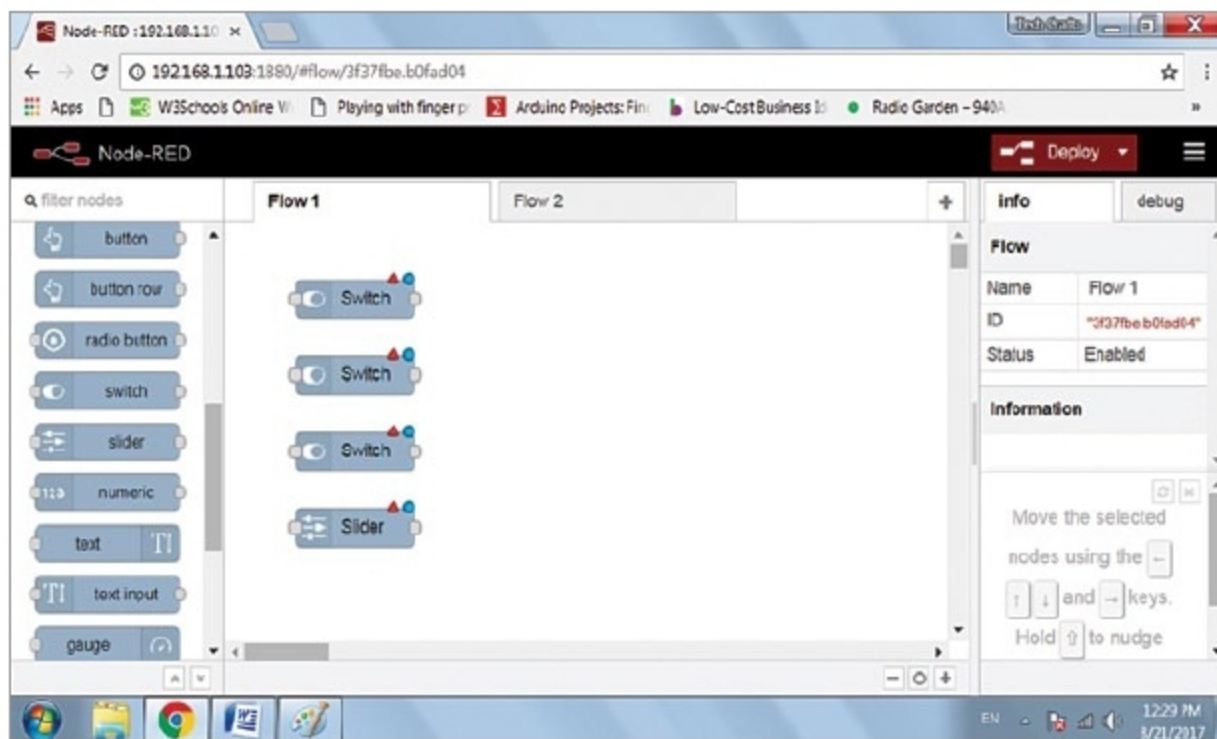


Fig. 3: Selection of nodes

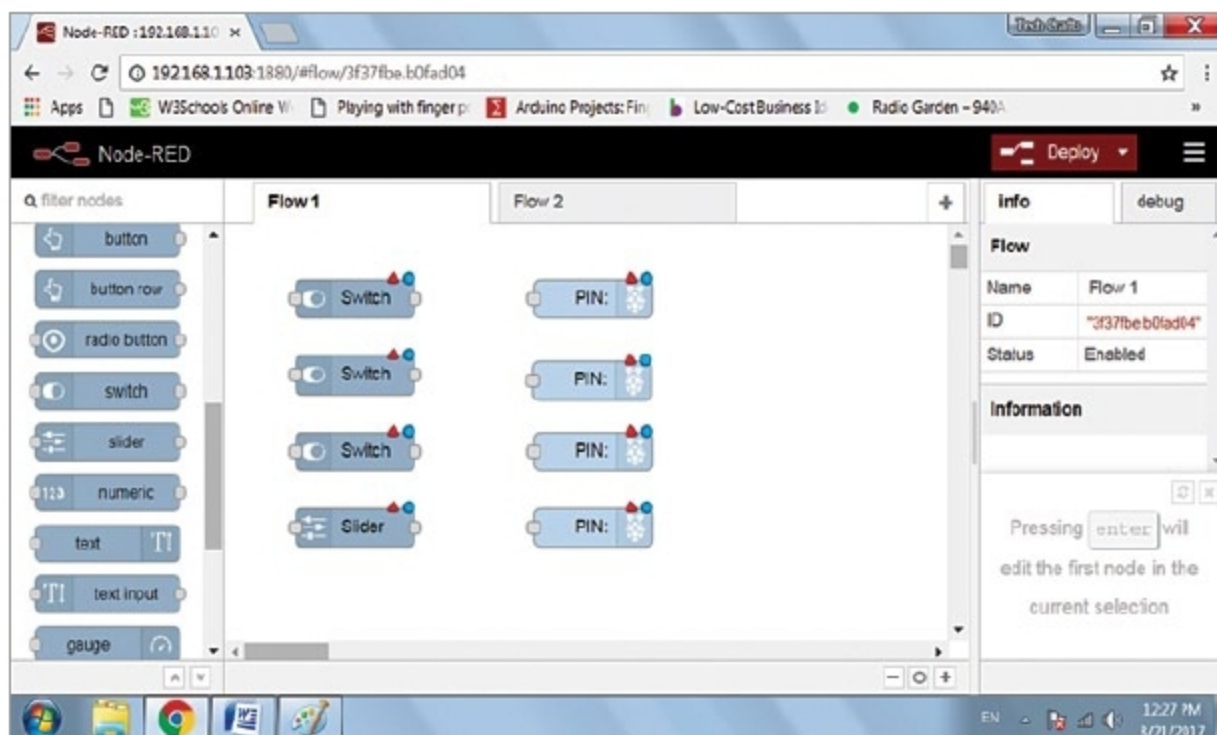


Fig. 4: Selection of pins

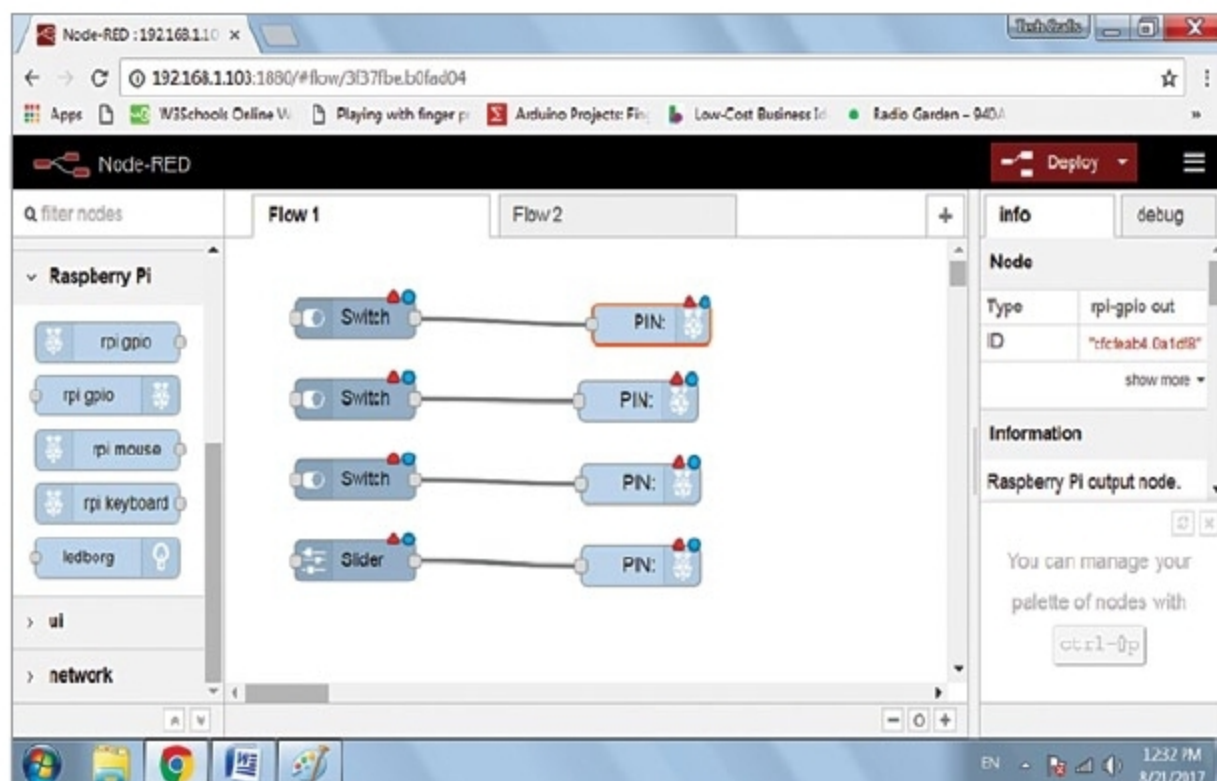


Fig. 5: Connection of nodes and GPIO

```
node-red-start // (to start Node-RED)
node-red-stop // (to stop Node-RED)
```

Node-RED console window will open, as shown in Fig. 1.

In case you do not know the IP address of your Raspberry Pi, type `ipconfig` in terminal window to check the IP address. Default port used in Node-RED for communication is 1880.

To open Node-RED graphical application, open the Web browser on Raspberry Pi and type the IP address of your Raspberry Pi as `https://192.168.1.103:1880`

You will get Node-RED graphical homepage, as shown in Fig. 2.

There are various coloured blocks on the left side. By scrolling down, you will see many nodes grouped together by categories including input, output, function, social, Raspberry Pi, UI and so on. You can use these nodes in your project by dragging and dropping them in the work area sheet at the centre. Different nodes are used for different operations.

GPIO and UI nodes may be missing in some older Node-RED versions. In that case, download these from the Internet by entering the following commands:

```
cd ~/.node-red
npm install node-red-contrib-gpio
npm install node-red-contrib-ui
```

Stop and restart Node-RED. You will get the updated nodes. (In some cases, rebooting the system may be required.)

Designing of home automation

The above application will be used to control four household loads/appliances such as fan, TV, tubelight and LED dimmer through a cell-phone. To do this, let us start with Node-RED UI node. We have a total of four loads—three switches and a slider—for four different loads. Drag and drop the switch and slider nodes from the node list. Do this four times for four different loads.

Note that, electrical loads will be controlled through UI switches and relays, as shown in the main circuit (Fig. 9). Relays are connected to GPIO pins of Raspberry Pi. Drag and drop four pins in the UI for Rasp-